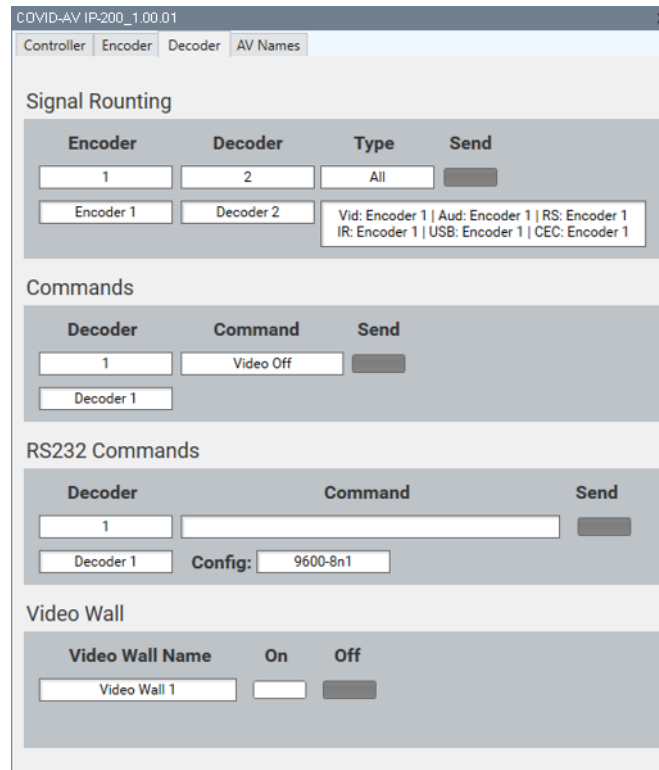
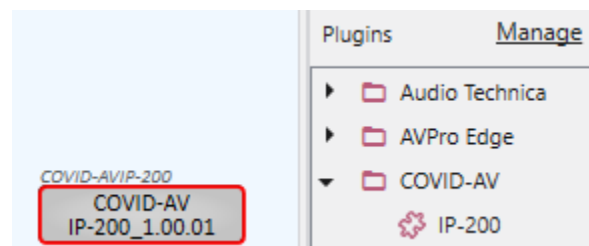


The COVID-AV IP-200 Q-SYS Plugin makes the operation of the IP-200 system very easy when building the Q-SYS User Interface design. Control of the IP-200 system is made through the CTRL-100 controller using TCP/IP or RS232. Programming can be done using the control pins for simple interfacing or for advanced users by using Component Controls.



To install the COVID-AV IP-200 Q-SYS Plugin, just double-click on the Plugin or place it in your documents folder under QSC\Q-Sys Designer\Plugins. When you open Q-SYS Designer you will see it in Plugins.



Properties

This Plugin dynamically configures its inputs and outputs to the size you set in the properties.

First, you need to set up the properties of the Plugin. The properties items include:

- Connection - TCP or Serial
- IP – The IP Address
- PresetSize – The number of presets you are using.
- EncoderSize - The number of Encoders you are using.
- DecoderSize - The number of Decoders you are using.

Properties	
IP-200 Properties	
Connection	TCP ▼
IP	192.168.10.173
TcpPort	4001
PresetSize	4
EncoderSize	2
DecoderSize	2
Show Debug	No ▼
Graphic Properties	
Label	COVID-AV IP-200_1.00.01
Position	1015,1002
Fill	
Script Access	
Code Name	COVID-AVIP-200
Script Access	None ▼

Control Pins

For easy configuration of your User Control Interface, the Plugin uses Control Pins. These Control Pins are dynamically set by the size you set in properties.

The following Control Pins are used:

- Audio Decoders
 - These pins are used to route audio from an Encoder to a Decoder using the String value of the Encoder.
- CEC Cmd Encoders
 - These Pins are used to send defined CEC receiver commands as a string.
- CMD Decoders
 - These Pins are used to send defined Decoder commands as a string.
- Preset Btns
 - Select a preset as a Boolean button command.
- Video Decoders
 - These pins are used to route video from an Encoder to a Decoder using the String value of the Encoder.
- API Direct
 - Use this pin to send a direct API command. Do not include the ">" or the Delimiter.

Properties	
IP-200 Properties	
Connection	TCP ▼
IP	192.168.10.173
TcpPort	4001
PresetSize	4
EncoderSize	2
DecoderSize	2
Show Debug	No ▼
Graphic Properties	
Label	COVID-AV IP-200_1.00.01
Position	1015,1002
Fill	
Script Access	
Code Name	COVID-AVIP-200
Script Access	None ▼
Control Pins ☐	
▲ Audio Decoders 1 ☐ 2 ☐ ▲ CEC Cmd Encoders 1 ☐ 2 ☐ ▲ Cmd Decoders 1 ☐ 2 ☐ ▲ Preset Btns 1 ☐ 2 ☐ 3 ☐ 4 ☐ ▲ Video Decoders 1 ☐ 2 ☐ API Direct ☐ Disable ☐	

Operation

The IP-200 Plugin will emulate the control of the system when it is not connected. This allows for testing of the User Control Interface. When connected the Network Connect Status LED will be green.

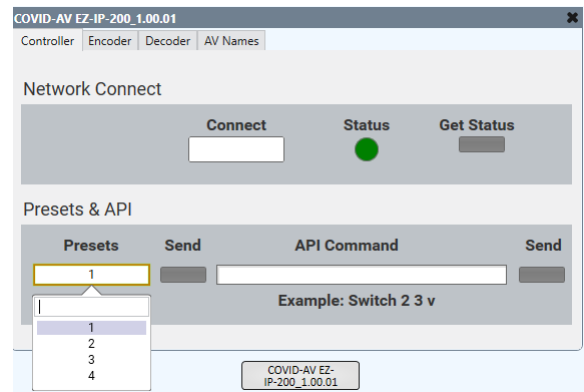
There are four pages associated with the Plugin:

1. Controller
2. Encoder
3. Decoder
4. Status

Controller Page

Network Connect

Although the IP-200 Plugin automatically connects at startup, the Connect button will allow you to Disconnect and Connect to the CTRL-100. A Green Status LED indicates that the Plugin is connected to the system. Once connected, the Plugin requests the status from the CTRL-100 controller. This status includes the routing information and the names of the Encoder and Decoders. If needed, the status can be requested by pressing the Get Status button.



Preset & API

One of the very powerful features of the IP-200 system is the Preset controls. Within the system up to 100 presets can be created to route AV and control displays.

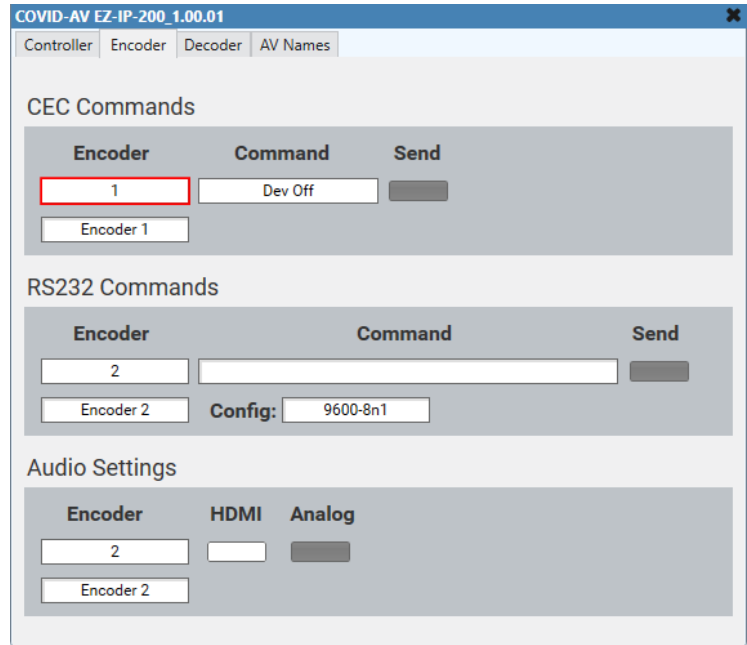
In some situations, you may want to send a direct API command to the IP-200 system. The manual lists all the commands. Just enter the command without the ">" and termination.

Encoder Page

This page allows the following:

- Sending selected CEC receiver commands to an Encoder
- Direct RS232 commands to an Encoder
- Setting the audio input of an Encoder

The name under the Encoder selection displays the name of the Encoder in the IP-200 system.



COVID-AV EZ-IP-200_1.00.01

Controller Encoder Decoder AV Names

CEC Commands

Encoder	Command	Send
1	Dev Off	Send
Encoder 1		

RS232 Commands

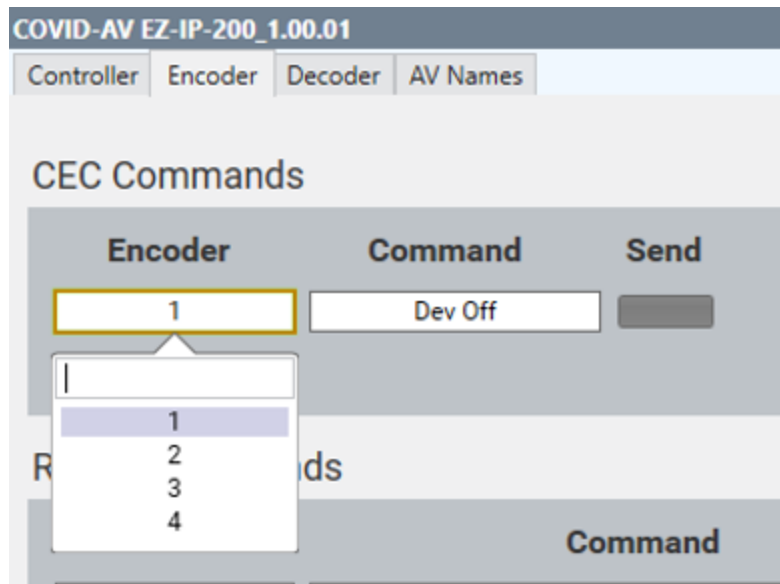
Encoder	Command	Send
2		Send
Encoder 2		
Config: 9600-8n1		

Audio Settings

Encoder	HDMI	Analog
2	☒	☐
Encoder 2		

CEC Commands

This page allows sending selected CEC receiver commands to an encoder to control a Blu-ray and ATV device. First select the Encoder followed by the command.



COVID-AV EZ-IP-200_1.00.01

Controller Encoder Decoder AV Names

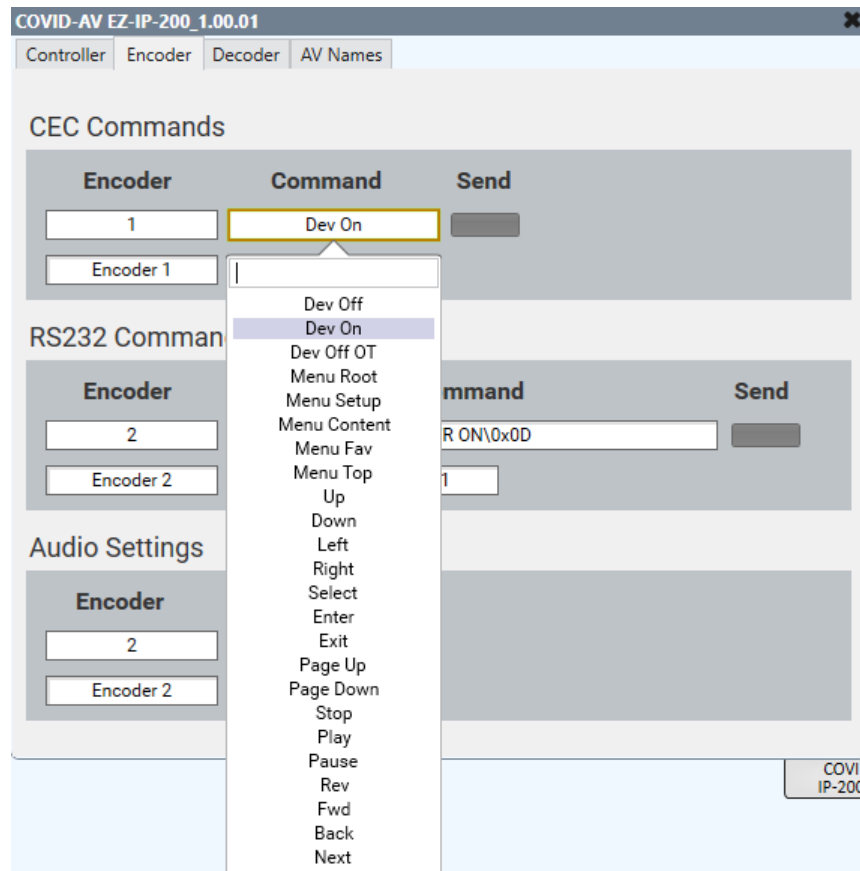
CEC Commands

Encoder	Command	Send
1	Dev Off	Send

Encoder 1

Command

1
2
3
4



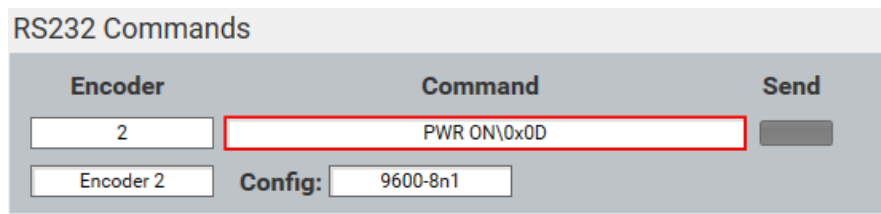
Encoder CEC Commands

RS232 Commands

RS232 Commands can be sent directly to the RS232 port on the Encoder. For UCI operation it is best to use the API Direct Control Pin. Example send to Encoder ID 2 “Power On\x0D\x0A”: “RS232E 2 9600-8n1 Power On\x0D\x0A”. Direct commands do not provide feedback.

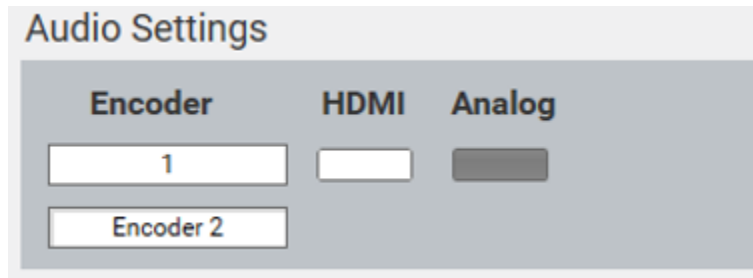
To send a command:

- Select the Encoder
- Enter the baud, data bits, parity and stop bits
- Enter the device command using String commands and “\0x##” for hex numbers
- Press Send



Audio Settings

The audio settings allow for selecting the audio type for the Encoder. The selection can be standard HDMI or Embed Analog audio.

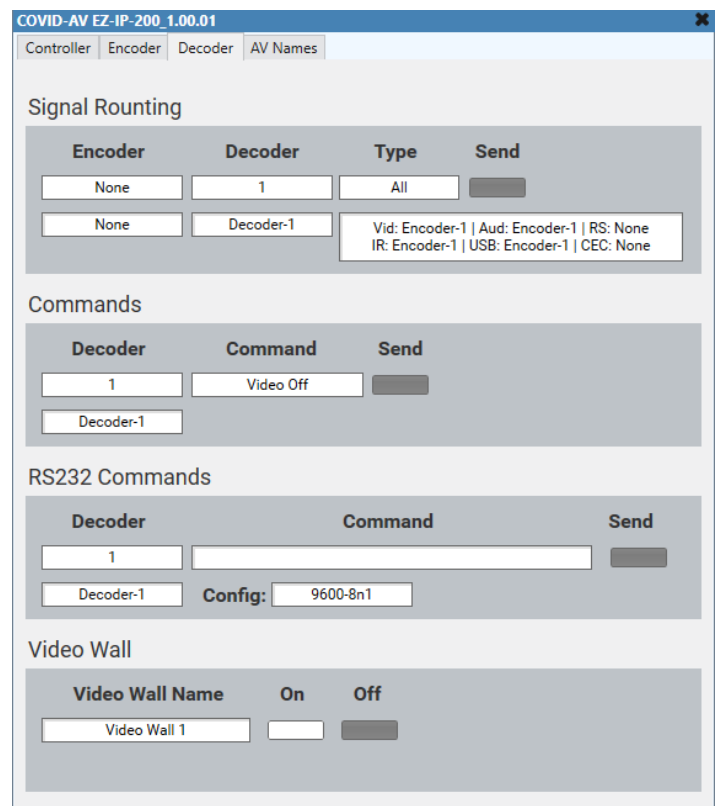


The screenshot shows the 'Audio Settings' window. It has three main sections: 'Encoder', 'HDMI', and 'Analog'. Under 'Encoder', there are two input fields: the first contains '1' and the second is labeled 'Encoder 2'. The 'HDMI' section has an empty input field. The 'Analog' section has a greyed-out button.

Decoder Page

The Decoder Page allows for the following:

- Signal Routing
- Commands
- RS232 Commands
- Video Wall

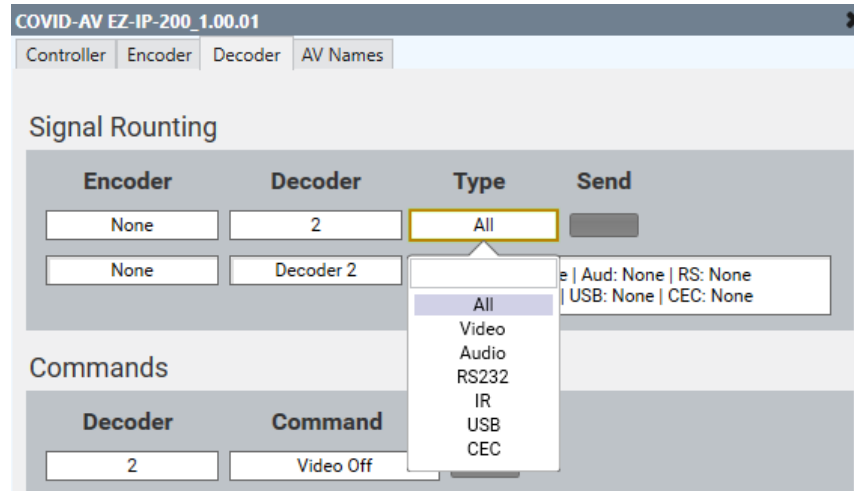


The screenshot shows the 'COVID-AV EZ-IP-200 1.00.01' window with the 'Decoder' tab selected. It contains four sections:

- Signal Routing:** A table with columns 'Encoder', 'Decoder', 'Type', and 'Send'. The first row has 'None', '1', 'All', and a button. The second row has 'None', 'Decoder-1', and a text box containing 'Vid: Encoder-1 | Aud: Encoder-1 | RS: None | IR: Encoder-1 | USB: Encoder-1 | CEC: None'.
- Commands:** A table with columns 'Decoder', 'Command', and 'Send'. The first row has '1', 'Video Off', and a button. The second row has 'Decoder-1'.
- RS232 Commands:** A table with columns 'Decoder', 'Command', and 'Send'. The first row has '1', an empty text box, and a button. The second row has 'Decoder-1', 'Config:', and a text box containing '9600-8n1'.
- Video Wall:** A table with columns 'Video Wall Name', 'On', and 'Off'. The first row has 'Video Wall 1', an empty input field, and a button.

Signal Routing

Routing is very simple by just selecting the Encoder ID followed by where you want it to go by the Decoder ID, the type of routing and then the Send button. A status box shows the current route for the Decoder. If the Encoder is set to None, the Route will be cleared in the Decoder.

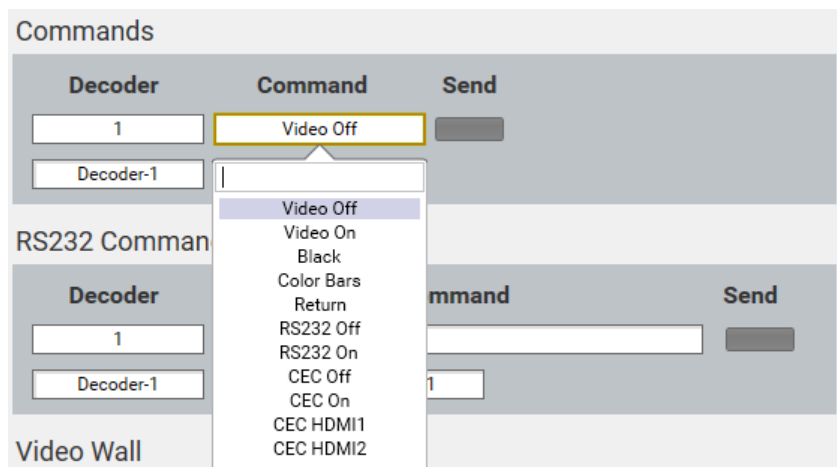


Commands

A list of commands can be selected from to send to a selected Decoder.

Commands:

- Video Off and On – Used to disable and enable the video output of the Decoder. This can be used to power on and off monitors that do not have control but can power by signal detection.
- Black, Color Bar and Return – These can be used for testing. Black can be used for picture blanking. To return to video, select Return.
- RS232 Off and On – Select these to send the RS232 power commands set in the Decoder.
- CEC Off and On – Select these to send the CEC power commands set in the Decoder.
- CEC HDMI 1&2 – These commands send the CEC input command of a display.

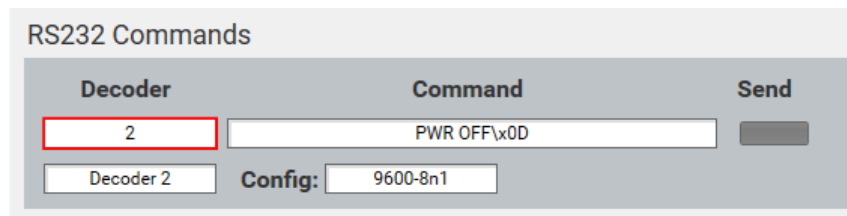


RS232 Commands

RS232 Commands can be sent directly to the RS232 port on the Encoder. For UCI operation it is best to use the API Direct Control Pin. Example send to Decoder ID 10 “Power On\x0D\x0A”:
“RS232D 10 9600-8n1 Power On\x0D\x0A”. Direct commands do not provide feedback.

To send a command:

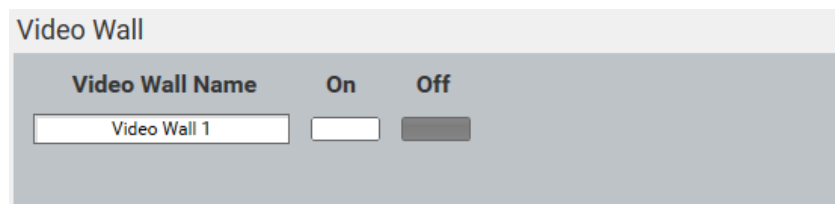
- Select the Encoder
- Enter the baud, data bits, parity and stop bits
- Enter the device command using String commands and “\0x##” for hex numbers
- Press Send



The screenshot shows the 'RS232 Commands' window. It has a header bar with the title 'RS232 Commands'. Below the header, there are three main sections: 'Decoder', 'Command', and 'Send'. The 'Decoder' section contains a text box with the value '2' and a button labeled 'Decoder 2'. The 'Command' section contains a text box with the value 'PWR OFF\x0D' and a button labeled 'Send'. The 'Send' button is disabled. Below the 'Decoder' and 'Command' sections, there is a 'Config:' label followed by a text box containing '9600-8n1'.

Video Wall

The operation of the Video Wall can be turned on or off. When on, the Video Wall is operational, and the Decoders are scaled to create a single image. When off, the Video Wall is disabled, and the Decoders are set to Single View allowing multiple displays like a quad view. Use the name assigned in the IP-200 system.



The screenshot shows the 'Video Wall' window. It has a header bar with the title 'Video Wall'. Below the header, there are three main sections: 'Video Wall Name', 'On', and 'Off'. The 'Video Wall Name' section contains a text box with the value 'Video Wall 1'. The 'On' section contains a button that is currently disabled. The 'Off' section contains a button that is currently active.

AV Names Page

This page allows you to monitor the routing of a Decoder. Select None for no status information. These can be used in the UCI to monitor what is sent to an output.

COVID-AV EZ-IP-200_1.00.01

Controller Encoder Decoder AV Names

Video Status - Select a Decoder

Status-1	Status-2	Status-3	Status-4
1	2	None	None
Encoder 3			
None	Status-6	Status-7	Status-8
1	None	None	None
2			
3			
4			

Audio Status - Select a Decoder

Status-1	Status-2	Status-3	Status-4
1	None	None	None
Encoder 2			

Video Status

There are up to 8 Decoders that can be monitored for their video routing.

Audio Status

There are up to 4 Decoders that can be monitored for their audio routing.

Control Pins

Control Pins are used to make it easy to interface the Plugin with the UCI.

Audio Decoders

These pins are used to route audio from an Encoder to a Decoder using the String value of the Encoder.

Sending Encoder 1 Audio to Decoder 1.

Using labels on a Selector connected to the Audio Decoders you can enter the Encoder ID, and this will route that Encoder to the Audio Decoder. Entering "0" clears the Audio Decoder. In Signal Routing Aud. Encoder 1 is shown, and Audio Status-1 set to Decoder 1, Encoder 1 is shown on the AV Name Page.

COVID-AV EZ-IP-200_1.00.01

Controller Encoder Decoder AV Names

Signal Routing

Encoder	Decoder	Type	Send
1	1	Audio	Send
		Vid: None Aud: Encoder 1 RS: None IR: None USB: None CEC: None	

Commands

Decoder	Command	Send
1	Video Off	Send

RS232 Commands

Decoder	Command	Send
1		Send
	Config: 9600-8n1	

Video Wall

Decoder	Decoders	On	Off
1	1	On	Off

Selector

- ☐ Label 1
- ☐ Label 2
- ☐ Label 3
- ☐ Label 4
- ☐ Output

COVID-AV EZ-IP-200_1.00.01

- ☒ Audio Decoders 1
- ☒ Audio Decoders 2
- ☒ Audio Decoders 3
- ☒ Audio Decoders 4

Selector

Selection	Label	Output	Sel	Selection	Output
1	1	Value 1	Sel		Output
2	2	Value 2	Sel		
3	3	Value 3	Sel		
4	4	Value 4	Sel		

Audio Status - Select a Decoder

Status-1	Status-2	Status-3	Status-4
1	None	None	None
Encoder 1			

CEC Cmd Encoders

These Pins are used to send defined CEC receiver commands as a string.

List of CEC Commands:

- Dev Off
- Dev Off
- Dev Off OT
- Menu Root
- Menu Setup
- Menu Content
- Menu Fav
- Menu Top
- Up
- Down
- Left
- Right
- Select
- Enter
- Exit
- Page Up
- Page Down
- Stop
- Play
- Pause
- Rev
- Fwd
- Back
- Next

COVID-AV EZ-IP-200_1.00.01

Controller

Encoder

Decoder

AV Names

CEC Commands

Encoder	Command	Send
1	Stop	Send
		Send

RS232 Commands

Encoder	Command	Send
1		Send
	Config: 9600-8n1	Send

Audio Settings

Encoder	HDMI	Analog
1	Send	Send
	Send	Send

Selector

COVID-AV EZ-IP-200_1.00.01

☐ Label 1
☐ Label 2
☐ Label 3
☐ Label 4
☐ Output

☒ CEC Cmd Encoders 1
☒ CEC Cmd Encoders 2
☒ CEC Cmd Encoders 3
☒ CEC Cmd Encoders 4

Selector

Selection	Label	Output	Sel	Selection	Output
1	Stop	Value 1	Send		Send
2	Play	Value 2	Send		
3	Pause	Value 3	Send		
4	Menu Setup	Value 4	Send		

CMD Decoders

These Pins are used to send defined Decoder commands as a string.

List of Decoder Commands:

- Video Off
- Video On
- Black
- Color Bars
- Return
- RS232 Off
- RS232 On
- CEC Off
- CEC On
- CEC HDMI1
- CEC HDMI2

Commands

Decoder	Command	Send
4	CEC On	Send
		Send

RS232 Commands

Decoder	Command	Send
1		Send
	Config: 9600-8n1	Send

Video Wall

Decoder	Decoders	On	Off
1	1	On	Off
		On	Off

Selector

☐ Label 1
☐ Label 2
☐ Label 3
☐ Label 4
☐ Output

COVID-AV EZ-IP-200_1.00.01

☒ cmd Decoders 1
☒ cmd Decoders 2
☒ cmd Decoders 3
☒ cmd Decoders 4

Selector

Selection	Label	Output	Sel	Selection	Output
1	Video Off	Value 1	Sel		Output
2	Video On	Value 2	Sel		
3	CEC Off	Value 3	Sel		
4	CEC On	Value 4	Sel		

Preset Btns

Select a preset as a Boolean button command, such as a selector.

COVID-AV EZ-IP-200_1.00.01

Controller

Encoder

Decoder

AV Names

Network Connect

Connect

Status

Get Status

Presets & API

Presets

Send

API Command

Send

2

Example: Switch 2 3 v

Selector

Output

Selector 1

Selector 2

Selector 3

Selector 4

COVID-AV EZ-IP-200_1.00.01

Presets 1

Presets 2

Presets 3

Presets 4

Selector

Selection	Label	Output	Sel	Selection	Output
1	1	Value 1		2	Value 2
2	2	Value 2			
3	3	Value 3			
4	4	Value 4			

Video Decoders

These pins are used to route video from an Encoder to a Decoder using the String value of the Encoder.

Using labels on a Selector connected to the Video Decoders you can enter the Encoder ID, and this will route that Encoder to the Video Decoder. Entering “0” or “None” clears the Video Decoder.

Sending Encoder 2 Video to Decoder 3.

In Signal Routing Vid. Encoder 2 is shown, and Video Status-1 set to Decoder 3, Encoder 2 is shown on the AV Name Page.

Signal Routing

Encoder	Decoder	Type	Send
2	1	Video	Send
Encoder-2	Decoder-1	Vid: Encoder-2 Aud: None RS: None IR: None USB: None CEC: None	

Selector
☐ Label 1
☐ Label 2
☐ Label 3
☐ Label 4
☐ Output

COVID-AV EZ-IP-200_1.00.01
☒ Video Decoders 1
☒ Video Decoders 2
☒ Video Decoders 3
☒ Video Decoders 4

Selector

Selection	Label	Output	Sel	Selection	Output
1	1	Value 1	Sel		
2	2	Value 2	Sel		
3	2	Value 3	Sel		
4	4	Value 4	Sel		

Video Status - Select a Decoder

Status-1	Status-2	Status-3	Status-4
3	None	None	None
Encoder 2			
Status-5	Status-6	Status-7	Status-8
None	None	None	None



COVID-AV IP-200 Q-SYS Plugin Version 1.00.01

API Direct

Use this pin to send a direct API command. Do not include the “>” or the Delimiter.

Command sent: “RS232D 1 9600-8n1 Power on\x0D\x0A”

All commands can be found the COVID AV EZ-200 manual.

COVID-AV EZ-IP-200_1.00.01

Controller Encoder Decoder AV Names

Network Connect

Connect Status Get Status

Presets & API

Presets Send API Command Send

1 RS232D 1 9600-8n1 Power on\x0D\x0A

Example: Switch 2 3 v

Selector

☐ Label 1 ☒ API Direct

☐ Output

Selector

Selection	Label	Output	Sel	Selection	Output
1	1 9600-8n1 Power on\	Value 1	☐	2	Value 2
2	2	Value 2	☐		
3	3	Value 3	☐		
4	4	Value 4	☐		